EISENSTEIN-RING STRUCTURE OF ALL ODD U-FAMILY VERTEX-SUPERALGEBRA SUPERCHARACTERS

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Allowed standard prerequisites (editorial).

- Standard q-binomial summation, vertex-algebra character definitions and classical elliptic/theta functions. Original source bibliography, including Lang, Arakawa and AM where relevant.

Source reference. Original reference 4.5: algebra-geometry/sources/sigma-2026-087/native/sigma26-087.tex, source lines 1594–1594. Cross-reference gives an introductory specialization, earlier narrative comparison or section pointer; all source-local arguments used for the counted theorem are included.

Source reference. Original reference 3.3: algebra-geometry/sources/sigma-2026-087/native/sigma26-087.tex, source lines 1209–1209. Cross-reference gives an introductory specialization, earlier narrative comparison or section pointer; all source-local arguments used for the counted theorem are included.

1. Multiple q-zeta values, quasi-modular forms, and characters

This section contains the second part of the paper. Throughout, we let $\tau \in \mathbb{H}$, $z \in \mathbb{C}$, $q = \mathrm{e}^{2\pi \mathrm{i}\tau}$, and $\zeta = \mathrm{e}^{2\pi \mathrm{i}z}$. Unless stated otherwise, all identities involving Fourier series in ζ are interpreted in the annulus specified in the statement, which determines the corresponding geometric-series expansion of each denominator. We begin by recalling the elliptic and quasi-Jacobi functions needed for our constant-term computations and then apply these techniques to characters of genus-zero class-$\mathcal{S}$ vertex operator algebras.

1.1. Elliptic functions, theta functions, and quasi-modular forms.

We present some basic (and mostly well-known) results on elliptic functions, (quasi)-Jacobi forms, and their Fourier coefficients. We closely follow [34].

The Jacobi theta function is given by

$$\vartheta(z;\tau) = \sum_{n \in \mathbb{Z}+\frac{1}{2}} \mathrm{e}^{\pi \mathrm{i} n^2 \tau + 2\pi \mathrm{i} n(z+\frac{1}{2})}.$$

We also define Eisenstein series with their q-expansion, for integers $k \geq 1$ (and with the usual convention that E_2 is quasimodular)

$$G_{2k}(\tau) = 2\zeta(2k) + \frac{(2\pi \mathrm{i})^{2k}}{(2k-1)!} \sum_{n \geq 1} \frac{n^{2k-1} q^n}{1-q^n}, \qquad q = \mathrm{e}^{2\pi \mathrm{i}\tau},$$

and slightly normalized versions

$$E_{2k}(\tau) = 1 - \frac{4k}{B_{2k}} \sum_{n \geq 1} \frac{n^{2k-1} q^n}{1-q^n}, \tag{1.1}$$

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where B_{2k} is the Bernoulli number. Recall that $\mathcal{QM}$ us the ring of quasi-modular forms for the full modular group discussed earlier. A famous result of S. Ramanujan is that

$$\left(q\frac{\mathrm{d}}{\mathrm{d}q}\right):\ \mathcal{QM}\to\mathcal{QM},$$

so $\mathcal{QM}$ is a differential ring. This well-known fact is employed throughout this section and in later parts.

The Weierstrass ζ-function[1] and the $\wp$-function are defined as

$$\zeta(z;\tau)=\frac{1}{z}+\sum_{(n,m)\neq(0,0)}\left(\frac{1}{z-m\tau-n}+\frac{1}{m\tau+n}+\frac{z}{(m\tau+n)^2}\right),$$
$$\wp(z;\tau)=\frac{1}{z^2}+\sum_{(n,m)\neq(0,0)}\frac{1}{(z-m-n\tau)^2}-\frac{1}{(m+n\tau)^2},$$
$$\wp'(z;\tau)=-\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}z}\wp(z;\tau).$$

Their Laurent expansions around $z=0$ are then given by

$$\zeta(z;\tau)=\frac{1}{z}-\sum_{n\geq 1}G_{2n+2}(\tau)z^{2n+1},$$
$$\wp(z;\tau)=\frac{1}{z^2}+\sum_{n\geq 1}(2n+1)G_{2n+2}(\tau)z^{2n},$$
$$\wp'(z;\tau)=\frac{1}{z^3}-\sum_{n\geq 1}n(2n+1)G_{2n+2}(\tau)z^{2n-1}.$$

For concrete computation, it is convenient to use the so-called P-functions or "propagators" (widely used for computations of genus one correlators in CFT and vertex operator algebra [52]):

$$P_k(\zeta;q)=\frac{(2\pi\mathrm{i})^k}{(k-1)!}\sum_{n\geq 1}\left(\frac{n^{k-1}\zeta^n q^n}{1-q^n}+(-1)^k\frac{n^{k-1}\zeta^{-n}}{1-q^n}\right).$$

Then inside $|q|<\left|\zeta^{-1}\right|<1$ we have

$$P_1\left(\mathrm{e}^{2\pi\mathrm{i}z},\tau\right)=-\zeta(z;\tau)+G_2(\tau)z-\pi\mathrm{i}, \tag{1.2}$$
$$P_2\left(\mathrm{e}^{2\pi\mathrm{i}z},\tau\right)=\wp(z,\tau)+G_2(\tau), \tag{1.3}$$
$$P_3\left(\mathrm{e}^{2\pi\mathrm{i}z},\tau\right)=-\wp'(z,\tau). \tag{1.4}$$

Similarly, $P_k\left(\mathrm{e}^{2\pi\mathrm{i}z};\tau\right)$, $k\geq 4$, is proportional to the $(k-2)$-th z-derivative of $\wp(z;\tau)$. The second and the third relations (1.3)–(1.4) are equivalent to the following:

$$\wp(z,\tau)=-G_2(\tau)+(2\pi\mathrm{i})^2\sum_{n\geq 1}\left(\frac{nq^n\zeta^n}{(1-q^n)}+\frac{n\zeta^{-n}}{(1-q^n)}\right),$$
$$\wp'(z,\tau)=-(2\pi\mathrm{i})^3\sum_{n\geq 1}\left(\frac{n^2q^n\zeta^n}{(1-q^n)}-\frac{n^2\zeta^{-n}}{(1-q^n)}\right).$$

To see that (1.2) holds, we require an identity

$$-\frac{1}{2}+\frac{\zeta}{\zeta-1}-\sum_{n\geq 1}\left(\frac{\zeta^{-1}q^n}{1-\zeta^{-1}q^n}-\frac{q^n\zeta}{1-q^n\zeta}\right)=\frac{1}{2}+\sum_{n\geq 1}\left(\frac{\zeta^n q^n}{1-q^n}-\frac{\zeta^{-n}}{1-q^n}\right), \tag{1.5}$$

[1]Not to be confused with the Fourier coefficient $\zeta=\mathrm{e}^{2\pi\mathrm{i}z}$ used below.

again valid in $|q| < |\zeta^{-1}| < 1$. We denote either side of (1.5) by $J_1(z;\tau)$. This function does not transform as a Jacobi form, instead we have

$$J_1(z + m\tau + n;\tau) = J_1(z;\tau) - m,$$
$$J_1\left(\frac{z}{\tau}; \frac{-1}{\tau}\right) = \tau J_1(z;\tau) + z.$$

We also require $\sigma(z;\tau)$, which can be defined using the theta function [34]

$$\sigma(z;\tau) = -\exp\left(\frac{\eta_1 z^2}{2}\right)\frac{\vartheta(z;\tau)}{2\pi\eta(\tau)^3},$$

where $\eta_1 = G_2(\tau)$ in the normalization of [34]; for our purposes, it is a function of τ only and will not be used elsewhere.

Functions $\wp$, ϑ, and σ are related through a fundamental relation [34]:

(1.6)
$$\wp(z_1;\tau) - \wp(z_2;\tau) = \frac{\sigma(z_1+z_2;\tau)\sigma(z_1-z_2;\tau)}{\sigma(z_1;\tau)^2\sigma(z_2;\tau)^2} = \frac{\vartheta(z_1+z_2;\tau)\vartheta(z_1-z_2;\tau)}{4\pi^2\vartheta(z_1;\tau)^2\vartheta(z_2;\tau)^2\eta(\tau)^6}.$$

Our next result concerns the constant term in the Fourier expansion of elliptic functions with respect to $\zeta = \mathrm{e}^{2\pi \mathrm{i} z}$. This result is well known and has appeared previously in the literature; see, for example, [26, 40]. In the present work, however, we adopt the approach of [14], where all Fourier coefficients of certain index-zero Jacobi forms are investigated, rather than focusing solely on the constant term.

Proposition 1.1. *Let $f(z;\tau) \in \mathbb{C}[\wp(z;\tau), \wp'(z;\tau)]$. Then, in the range $|q| < \left|\zeta^{-1}\right| < 1$, the constant term $\mathrm{CT}_\zeta f(z;\tau)$ belongs to $\mathcal{QM}$, and the non-constant terms $\mathrm{CT}_\zeta f(z;\tau)\zeta^{-r}$, $r \neq 0$, are not quasi-modular.*

Proof. We may assume that $f = \wp^r(\wp')^s$. Using the Weierstrass equation

$$(\wp')^2 = 4\wp^3 - g_2(\tau)\wp - g_3(\tau),$$

where $g_2(\tau)$ and $g_3(\tau)$ are multiples of $E_4(\tau)$ and $E_6(\tau)$, respectively, every higher power of $\wp'$ can be reduced to the cases $s = 0$ and $s = 1$. We use the Laurent expansion in the variable z around zero to write

$$f(z;\tau) = \sum_{0 \le i < \frac{\ell}{2}} \frac{D_{\ell-2i}(\tau)}{z^{\ell-2i}} + D_0(z) + O(z),$$

where the z-coefficients $D_{\ell-2i}(\tau)$ are holomorphic modular forms of weight $2i$ that can be explicitly computed using the Laurent expansion of $\wp$ and $\wp'$, and $\ell = 2r+3s$. Of course, $D_0(z)$ could be zero depending on the parity of s. To compute the ζ-constant term, we proceed as in [14, Theorem 1.1] dealing with a particular Jacobi form of index zero (the case when m is even in [14] is precisely the case of elliptic functions here).

For the nonzero Fourier coefficient $f_r(\tau)$, $r \neq 0$, we compute the contour integral

$$\int_{\partial P_{z_0}} f(w;\tau)\mathrm{e}^{-2\pi \mathrm{i} r w}\,\mathrm{d}w,$$

where

$$P_{z_0} = \{z_0 + r\tau + s \mid 0 \le r, s \le 1\},$$

in two different ways: directly as a contour integral, and by the residue theorem by choosing

$$z_0 = -\tfrac{1}{2} - \tfrac{\tau}{2},$$

so that the only pole inside the contour is at $z = 0$. This quickly leads to a formula for the r-th coefficient, which is quasi-modular if and only if $r = 0$ and not quasi-modular otherwise (see [14, Theorem 1.1]). More explicitly,

$$\mathrm{CT}_\zeta f(z;\tau) = D_0(\tau) + \sum_i \lambda_i D_{2i}(\tau) E_{2i}(\tau),$$

where λ_i are some coefficients. This is certainly contained in $\mathcal{QM}$. □

1.2. **Quasi-Jacobi forms.** We are also interested in Fourier coefficients, or more precisely the constant term, of more general functions that include τ-derivatives of elliptic functions. For that we need the concept of *quasi-Jacobi forms* (see [37, 42, 50] for instance). We are only concerned with the subspace of meromorphic quasi-Jacobi forms of index 0 with poles at $L_\tau = \mathbb{Z}\tau + \mathbb{Z}$. These are known to form a subring $\mathcal{QJ}^0 := \mathbb{C}[E_6, E_4, J_1, \wp, \wp']$ inside the ring of quasi-Jacobi forms $\mathcal{QJ} := \mathbb{C}[F, E_6, E_4, J_1, \wp, \wp']$ (E_6 can be omitted from both rings though), where F is a particular quasi-Jacobi form of non-zero index not needed here [42]. Besides $J_1(z;\tau)$ mentioned earlier, we also require more general quasi-Jacobi forms ($k \geq 2$):

$$J_k(z;\tau) = B_k - k \sum_{n,r\geq 1} r^{k-1}\big(\zeta^n + (-1)^k \zeta^{-n}\big) q^{nr}.$$

Next we define the polynomials $p_m(x)$, $m \geq 2$ via

$$\sum_{n\geq 1} n^{m-1} x^n = \frac{x p_m(x)}{(1-x)^m}.$$

For instance, $p_2(x) = 1$, $p_3(x) = 1 + x$, etc. These are known as (slightly shifted) Euler's polynomials. The next result is elementary and essentially follows from the definition of $p_m(x)$.

Lemma 1.2. *We have*

(1) *For every even $s \geq 0$, and $k \geq 1$,*

$$\sum_{r\geq 1} \frac{r^s q^r p_{2k}(q^r)}{(1-q^r)^{2k}} \in \mathcal{QM}.$$

(2) *For every odd $s \geq 1$, and $k \geq 1$,*

$$\sum_{r\geq 1} \frac{r^s q^r (1+q^r) p_{2k}(q^r)}{(1-q^r)^{2k+1}} \in \mathcal{QM}.$$

Proof. For (1), it is enough to observe:

$$\sum_{r\geq 1} r^j \frac{q^r p_{a+j}(q^r)}{(1-q^r)^{a+j}} = \sum_{r\geq 1,\, n\geq 1} r^j n^{a+j-1} q^{nr} = \left(q\frac{\mathrm{d}}{\mathrm{d}q}\right)^j \sum_{r\geq 1,\, n\geq 1} n^{a-1} q^{nr} = \left(q\frac{\mathrm{d}}{\mathrm{d}q}\right)^j \sum_{n\geq 1} \frac{n^{a-1} q^n}{1-q^n}.$$

This implies that, for every s and k as above, we may choose $a \geq 2$ and $j \geq 0$ such that the sum in (1) is a (higher) derivative of an Eisenstein series, or an Eisenstein series with its constant term removed. Similarly, any series appearing in (2) can be expressed as a linear combination of derivatives of two Eisenstein series. In either case, the resulting sum lies in $\mathcal{QM}$. □

Theorem 1.3. *Let $f(z;\tau) \in \mathcal{QJ}^0$. Then in the range $|q| < |\zeta^{-1}| < 1$, $\mathrm{CT}_\zeta f(z;\tau) \in \mathcal{QM}$.*

Proof. Compared with Proposition 1.1, where f is doubly periodic in z and belongs to $\mathbb{C}[\wp, \wp']$, the only new issue is the contribution of $J_1(z;\tau)$ and thus it suffices to prove the statement for

$$f(z;\tau) = J_1(z;\tau)^j g(z;\tau),$$

where $g(z;\tau) \in \mathbb{C}[\wp, \wp']$ is elliptic and $j \geq 1$. We use induction on j. For $j = 1$, we have to analyze

$$J_1(z;\tau)g(z;\tau),$$

where $g(z;\tau)$ is elliptic. We will compute the constant term by expanding $g(z;\tau) = \zeta^r g_r(\tau)$, where $g_0(\tau)$ is quasi-modular (possibly zero) and non-zero Fourier coefficients are given by [14]

$$g_r(\tau) = \frac{q^r}{1-q^r} \sum_{0 \leq \ell < m/2} r^{m-2\ell-1} D_\ell(\tau), \qquad r > 0,$$

and

$$g_r(\tau) = -\frac{1}{1-q^{-r}} \sum_{0 \leq \ell < m/2} r^{m-2\ell-1} D_\ell(\tau), \qquad r < 0,$$

where the parity of m depends on the parity of the z powers in the Laurent expansion of $g(z;\tau)$, and $D_\ell(\tau)$ are modular forms as in Proposition 1.1. Multiplying out $J_1(z;\tau) = \frac{1}{2} + \sum_{n \geq 1} \left(\frac{\zeta^n q^n}{1-q^n} - \frac{\zeta^{-n}}{1-q^n} \right)$ and $g(z;\tau)$, we quickly find the constant term of $J_1(z;\tau)g(z;\tau)$ to be

$$\frac{1}{2} g_0(\tau) - \sum_{0 \leq \ell < m/2} D_\ell(\tau) \sum_{r \geq 1} \frac{r^{m-2\ell-1} q^r}{(1-q^r)^2} - \sum_{0 \leq \ell < m/2} D_\ell(\tau) \sum_{r \geq 1} \frac{(-r)^{m-2\ell-1} q^r}{(1-q^r)^2}.$$

If m is even, then the two sums cancel out and we end up with $\frac{1}{2} g_0(\tau)$ for the constant term which is certainly in $\mathcal{QM}$ by Proposition 1.1. If m is odd, then we get a contribution from the sum

$$\sum_{r \geq 1} \frac{r^s q^r}{(1-q^r)^2},$$

where $s := m - 2\ell - 1 \geq 0$ is even. For $s = 0$, this sum equals $\frac{1}{24}(1 - E_2(\tau)) \in \mathcal{QM}$. For $s \geq 2$, the sum is the first $\left(q \frac{\mathrm{d}}{\mathrm{d}q}\right)$-derivative of the Eisenstein series $E_{s-1}(\tau)$ and thus in $\mathcal{QM}$.

For the induction step, we need [41, formula (11)] that allows us to express J_1^ℓ as a linear combination of J_ℓ, an elliptic Jacobi form K_ℓ, and quasi-Jacobi forms involving J_1^k, $k < \ell$, for which the induction hypothesis applies. Thus in order to finish the proof we have to argue that

$$\mathrm{CT}_\zeta J_\ell(z;\tau) g(z;\tau) \in \mathcal{QM},$$

where $g(z;\tau)$ is elliptic. As in the $j = 1$ case, we first observe that

$$\frac{1}{\ell}(B_\ell - J_\ell(z;\tau)) = \sum_{n \geq 1} \left(\frac{q^n p_\ell(q^n) \zeta^n}{(1-q^n)^\ell} + (-1)^\ell \frac{q^n p_\ell(q^n) \zeta^{-n}}{(1-q^n)^\ell} \right),$$

where p_ℓ are as above. We distinguish two cases: ℓ is even and ℓ is odd. Assume first that $\ell = 2k$ is even. After we multiply out Fourier expansions and extract the constant term as above, the constant term takes the shape

$$\sum_{r \geq 1} \frac{q^{2r} r^s p_{2k}(q^r)}{(1-q^r)^{2k+1}} - (-1)^s \sum_{r \geq 1} \frac{q^r r^s p_{2k}(q^r)}{(1-q^r)^{2k+1}},$$

where $s := m - 2n - 1$ could be either even or odd. For s even, we have

$$\sum_{r \geq 1} \frac{r^s q^{2r} p_{2k}(q^r)}{(1-q^r)^{2k+1}} - (-1)^s \sum_{r \geq 1} \frac{r^s q^r p_{2k}(q^r)}{(1-q^r)^{2k+1}} = -\sum_{r \geq 1} \frac{r^s q^r p_{2k}(q^r)}{(1-q^r)^{2k}}$$

whereas for s odd, we obtain

$$\sum_{r\geq 1}\frac{r^s q^{2r}p_{2k}(q^r)}{(1-q^r)^{2k+1}}+\sum_{r\geq 1}\frac{r^s q^{r}p_{2k}(q^r)}{(1-q^r)^{2k+1}}=\sum_{r\geq 1}\frac{r^s q^{r}(1+q^r)p_{2k}(q^r)}{(1-q^r)^{2k+1}}.$$

By Lemma 1.2, both series are in $\mathcal{QM}$. Similarly we argue for ℓ odd. That completes the proof. □

Denote by $D_z=\frac{1}{2\pi \mathrm{i}}\frac{\partial}{\partial z}$ and $D_\tau=\frac{1}{2\pi \mathrm{i}}\frac{\partial}{\partial \tau}$, and let $s(x,y)\in\mathbb{C}[x,y]$ be a polynomial with complex coefficients.

Corollary 1.4. *Let $f(z;\tau)\in\mathcal{QJ}^0$. Then*

$$\mathrm{CT}_\zeta(s(D_z,D_\tau)f(z;\tau))\in\mathcal{QM}.$$

Proof. We only have to argue that the ring $\mathcal{QJ}^0$ is closed under D_z and D_τ. This can be verified explicitly on the generators J_1, $\wp$, $\wp'$, E_4, E_6 (see, for instance, [42, 50]). □

An important special case involves products of derivatives of the P-functions introduced earlier. A similar result appeared previously in [26, Proposition 5.9], though our proof differs.

Proposition 1.5. *We have*

(*i*) *For $m\geq 1$ and $k\geq 1$,*

$$\mathrm{CT}_\zeta\big(P_k(\zeta;\tau)^m\big)\in\mathcal{QM}.$$

(*ii*) *More generally, for every polynomial $s(x,y)$ and $k\geq 1$,*

$$\mathrm{CT}_\zeta\big\{(s(D_\tau,D_z)P_k(\zeta;\tau))^m\big\}\in\mathcal{QM}.$$

Proof. Part (ii) follows immediately from Theorem 1.3, Corollary 1.4, and formulas (1.2)–(1.4). □

2. Characters and supercharacters of the $\mathcal{U}$ family of vertex operator (super)algebras

In [1], Adamović and the author introduced an infinite family of vertex operator (super)algebras, somewhat analogous to those appearing in the class-$\mathcal{S}$ and in the Cvitanović–Deligne series [6, 10]. We refer to this family as the $\mathcal{U}$ family of vertex operator (super)algebras and denote its members by $\mathcal{U}^{(m)}$. As demonstrated in [1], for each integer $m\geq 3$, $\mathcal{U}^{(m)}$ is constructed as an infinite (super)extension of the simple affine vertex operator algebra $V_{-1}(\mathfrak{sl}(m))$ of central charge $-(m+1)$.

When m is odd, $\mathcal{U}^{(m)}$ is a $\mathbb{Z}$-graded vertex operator superalgebra, and it is therefore natural to consider its supercharacter $\mathrm{sch}[\mathcal{U}^{(m)}](\tau)$. Recall that the *supercharacter* of a vertex operator superalgebra is defined by taking the difference of the graded characters of its even and odd subspaces. Equivalently, it is obtained by replacing the ordinary trace in the definition of the character with the supertrace. When m is even, $\mathcal{U}^{(m)}$ is a vertex operator algebra with $\frac{1}{2}\mathbb{Z}_{\geq 0}$-grading. For simplicity, we restrict our attention here to the case where m is odd; see Remark 4.5. The main motivation for the study undertaken in [1] was the following:

Conjecture 2.1 ([1])**.** *For every $m\geq 3$, the vertex operator (super)algebra $\mathcal{U}^{(m)}$ is quasi-lisse.*

One of the main results of [1] is that the characters and supercharacters of a distinguished family of $\mathcal{U}^{(m)}$-modules (including $\mathcal{U}^{(m)}$ itself) exhibit quasi-modular behavior. This observation provided one of the primary motivations for the conjecture.

In this section, we present two new results on supercharacters that refine and extend the results of [1]:

(1) We derive a more explicit formula in terms of the constant terms of powers of the Weierstrass elliptic function, in the spirit of the computations carried out in Section 3.3.
(2) We prove that the supercharacter can be expressed in terms of quasi-modular forms of a specific type on the congruence subgroup $\Gamma_0(m)$.

An important role in our construction is played by the index-zero Jacobi form

$$F_m(z;\tau) = \epsilon_m \eta(\tau)^{m+1} \frac{\vartheta(mz;m\tau)}{\vartheta(z;\tau)^m},$$

where ϵ_m is some non-zero constant defined in [1]. Then we have the following result.

Theorem 2.2 ([1])**.** *For m odd, we have (in a suitable ζ-range)*

$$\mathrm{CT}_\zeta F_m(z;\tau) = \mathrm{sch}[\mathcal{U}^{(m)}](\tau),$$

where as before $\zeta = \mathrm{e}^{2\pi \mathrm{i} z}$.

2.1. **Supercharacters of $\mathcal{U}^{(m)}$ via elliptic functions.** Using the notation from [34], let

$$G_{2,N,(0,\frac{r}{N})}(\tau) := \wp\left(\frac{r}{N};\tau\right)$$

with $1 \le r \le N-1$, a certain weight two Eisenstein series of level N. Then we have for $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{Z})$,

$$G_{2,N,(0,\frac{r}{N})}\left(\frac{a\tau+b}{c\tau+d}\right) = (c\tau+d)^2 G_{2,N,A(0,\frac{r}{N})}(\tau).$$

Consider now $c|N$, so that $\gcd(d,N)=1$. Then the right-hand side equals

$$(c\tau+d)^2 G_{2,N,(\frac{cr}{N},\frac{dr}{N})}(\tau) = (c\tau+d)^2 G_{2,N,(0,\frac{dr}{N})}(\tau).$$

In other words, elements of $\Gamma_0(N)$ permute the G_2-series. Using $\wp(-z;\tau) = \wp(z;\tau)$ and letting N be odd, we see that

$$\mathbb{G}_1(\tau) := \sum_{a=1}^{(N-1)/2} G_{2,N,(0,\frac{a}{N})}(\tau)$$

is invariant under $\Gamma_0(N)$. Similarly, we evaluate elementary symmetric polynomials $e_i(x_1,\dots,x_{(N-1)/2})$, $1 \le i \le (N-1)/2$, at $G_{2,N,(0,\frac{r}{N})}(\tau)$ and construct genuine modular forms on $\Gamma_0(N)$, which we denote by $\mathbb{G}_i(\tau)$ (observe that $\mathbb{G}_1(\tau)$ comes from $x_1 + x_2 + \cdots + x_{(N-1)/2}$). These are modular forms of weight $2i$.

Now we are ready to give a new formula for the supercharacter of $\mathcal{U}$ expressed using $\mathrm{CT}_\zeta \wp(z)^k$.

Proposition 2.3. *We have the supercharacter formula*

$$\mathrm{sch}[\mathcal{U}^{(m)}](\tau) = \mu_m \frac{\eta(m\tau)^3}{\eta(\tau)^{2m-1}} \sum_{k=0}^{(m-1)/2} (-1)^k \mathbb{G}_k(\tau) \mathrm{CT}_\zeta \wp(z;\tau)^k,$$

where $\mu_m \neq 0$ is a constant depending only on m.

Proof. We first recall identity (1.6) that allows us to write

$$\wp(z;\tau) - \wp\left(\frac{j}{m};\tau\right) = \frac{\theta\left(z+\frac{j}{m};\tau\right)\theta\left(z-\frac{j}{m};\tau\right)}{4\pi^2\theta(z;\tau)^2\theta\left(\frac{j}{m};\tau\right)^2\eta(\tau)^6}.$$

Multiplying out these relations from $j=1,\dots,\frac{m-1}{2}$ (here $m=2k+1$ is odd!), using

$$\big(1-x^{2k+1}\big)=(1-x)\prod_{j=1}^{k}\big(1-x\mathrm{e}^{\frac{2\pi j}{2k+1}}\big)\prod_{j=1}^{k}\big(1-x\mathrm{e}^{-\frac{2\pi j}{2k+1}}\big),$$

modulo Euler factors, we can express the Jacobi form $F_m(z;\tau)$ as the product of $\wp(\tau)-\wp\big(\frac{k}{m};\tau\big)$, where $1\leq k\leq (m-1)/2$. After collecting the z-independent theta and eta factors, put $r=(m-1)/2$ and $a_j=\wp(j/m;\tau)$. Vieta's formula gives

$$\prod_{j=1}^{r}(X-a_j)=\sum_{k=0}^{r}(-1)^{r-k}e_{r-k}(a_1,\dots,a_r)X^k.$$

Taking $X=\wp(z;\tau)$ shows explicitly that the coefficient of $\wp(z;\tau)^k$ is $(-1)^{r-k}e_{r-k}(a_1,\dots,a_r)$; these elementary symmetric functions are precisely the forms denoted by $\mathbb{G}_{r-k}(\tau)$, up to the common normalization absorbed into μ_m. □

As an application, we compute the supercharacter for $m=3$. We only have to examine $\wp(z;\tau)-\wp(\frac{1}{3},\tau)$. Explicit computation with Eisenstein series gives

$$\wp\left(\frac{1}{3};\tau\right)=\frac{3}{2}G_2(\tau)-\frac{9}{2}G_2(3\tau).$$

Combined with the fact that the constant term of $\wp(z;\tau)$ is $G_2(\tau)$, we obtain

$$G_2(\tau)-(\frac{3}{2}G_2(\tau)-\frac{9}{2}G_2(3\tau))=-\frac{1}{2}G_2(\tau)+\frac{9}{2}G_2(3\tau).$$

This agrees with the formula obtained in [1]:

$$\begin{aligned}\mathrm{sch}[\mathcal{U}^{(3)}](\tau)&=\frac{\eta^3(3\tau)}{\eta^5(\tau)}\left(-\frac{1}{8}E_2(\tau)+\frac{9}{8}E_2(3\tau)\right)\\&=q^{\frac{1}{6}}\big(1+8q+44q^2+152q^3+487q^4+1352q^5+\cdots\big).\end{aligned}$$

We should point out that $E_2(\tau)-9E_2(3\tau)$ is not modular on $\Gamma_0(3)$ but quasi-modular. More precisely, up to a scalar, it equals the logarithmic derivative of $\frac{\eta(3\tau)^3}{\eta(\tau)}$. Another slightly more involved example is $m=5$. Analyzing $\wp\big(\frac{i}{5};\tau\big)$, $1\leq i\leq 2$, gives

$$\begin{aligned}&\mathrm{sch}[\mathcal{U}^{(5)}](\tau)\\&=\frac{\eta^3(5\tau)}{\eta^9(\tau)}\left(\frac{25}{1152}E_2^2(\tau)-\frac{125}{192}E_2(\tau)E_2(5\tau)+\frac{3125}{1152}E_2(5\tau)^2+\frac{1}{576}E_4(\tau)-\frac{625}{576}E_4(5\tau)\right)\\&=q^{\frac{2}{3}}\big(1+24q+249q^2+1750q^3+9750q^4+45750q^5+\cdots\big).\end{aligned}$$

The appearance of the series $E_{2k}(m\tau)$ in these formulas is not an accident.

Theorem 2.4. *Assume $m\geq 3$ is odd. Then we have*

$$\mathrm{sch}[\mathcal{U}^{(m)}](\tau)=\frac{\eta(m\tau)^3}{\eta(\tau)^{2m-1}}\mathbb{F}_m(\tau),$$

where $\mathbb{F}_m(\tau)\in\mathbb{Q}[E_2(\tau),E_4(\tau),E_6(\tau),E_2(m\tau),E_4(m\tau),E_6(m\tau)]$ is a quasi-modular form on $\Gamma_0(m)$ of weight $m-1$.

Proof. Let $\mathcal{QM}(m):=\mathbb{Q}[E_2(\tau),E_4(\tau),E_6(\tau),E_2(m\tau),E_4(m\tau),E_6(m\tau)]$. We only have to find a Laurent expansion that involves suitable Eisenstein series. For this, we use a well-known identity [34]

$$\vartheta(z;\tau)=-2\pi z\eta(\tau)^3\cdot\exp\left(-\sum_{k=1}^{\infty}\frac{G_{2k}(\tau)}{2k}z^{2k}\right).$$

The Laurent coefficients in the expansion around $z = 0$ of

$$\frac{\vartheta(mz; m\tau)}{\vartheta(z;\tau)^m}$$

are of the form $E_{2i}(\tau)$ and $E_{2i}(m\tau)$. Consequently, they lie in the ring generated by $E_{2i}(\tau)$ and $E_{2i}(m\tau)$ for $i = 1, 2, 3$, and hence belong to $\mathcal{QM}(m)$.

Using the same argument as in Theorem 1.3 or in [1], we then show that the constant term can be expressed in terms of these Eisenstein series together with ordinary quasi-modular forms.

Finally, to see that $\mathbb{F}_m(\tau)$ is a quasi-modular form on the congruence subgroup $\Gamma_0(m)$, it suffices to observe that

$$E_2^{(m)}(\tau) := E_2(\tau) - mE_2(m\tau), \qquad E_4(m\tau), \qquad E_6(m\tau)$$

are modular forms on $\Gamma_0(m)$. Therefore, any homogeneous polynomial in

$$E_2(\tau),\ E_4(\tau),\ E_6(\tau),\ E_2(m\tau),\ E_4(m\tau),\ \text{and } E_6(m\tau)$$

is a quasi-modular form on $\Gamma_0(m)$. □

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